

NeuroFlow CFD Analyst

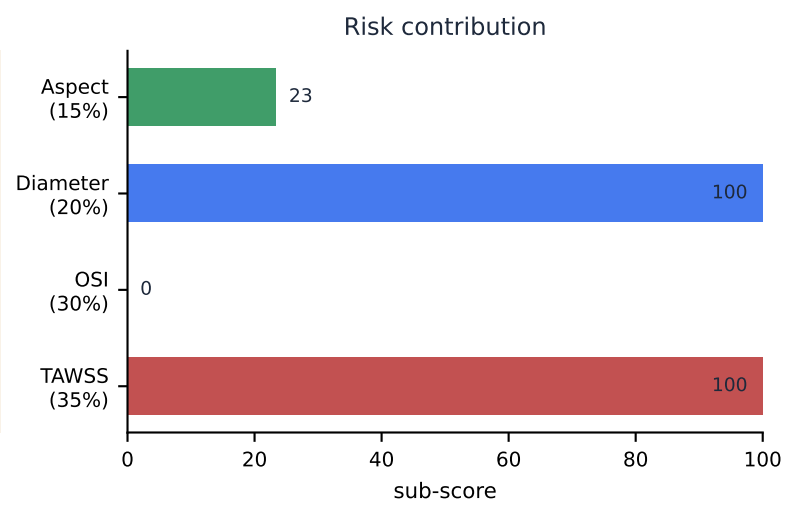
Clinical Hemodynamic Assessment

Case PT-2026-0102 · generated 2026-08-04 18:02

59

Composite Risk Index / 100

Moderate



Hemodynamic parameters

Parameter	Value	Threshold	Status
TAWSS — aneurysm dome	0.141 Pa	< 0.4 Pa	FLAGGED
TAWSS — parent artery	2.930 Pa	reference	normal
OSI — aneurysm dome	0.000	> 0.2	normal
RRT — aneurysm dome	21.35 Pa ⁻¹	> 3.0	FLAGGED
ECAP — aneurysm dome	0.000 Pa ⁻¹	> 1.0	normal
NWSS (sac / parent)	0.048	—	normal
LSAR (relative, <10% parent)	92.0 %	—	normal
LSAR (absolute, <0.4 Pa)	94.1 %	—	normal

Morphology (measured from the reconstructed surface)

Max diameter	Neck diameter	Aspect ratio	Dome-to-neck
11.02 mm	8.22 mm	1.12	1.34
Volume	Surface area	Non-sphericity	
665.6 mm ³	420.5 mm ²	0.304	

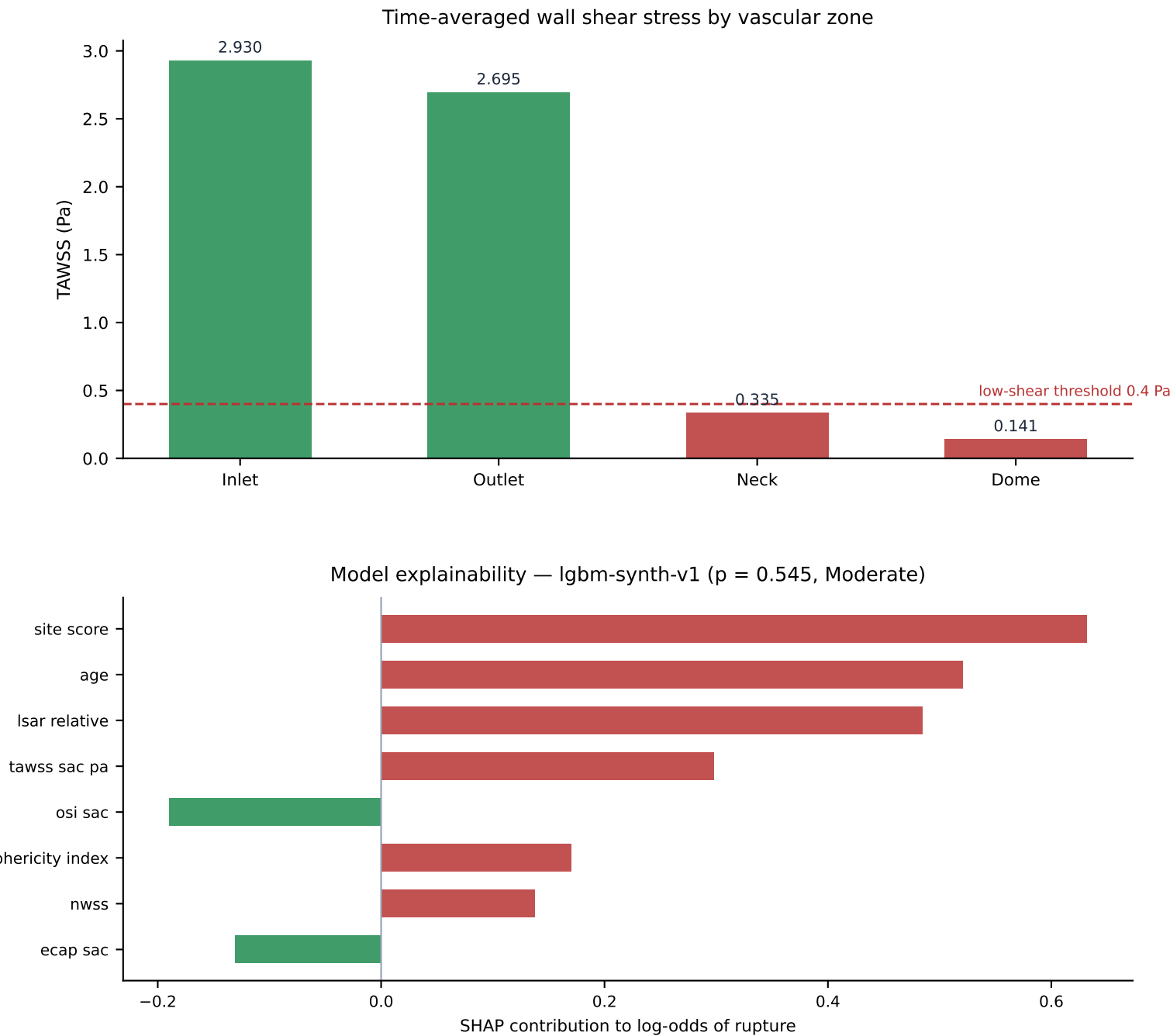
Provenance

Solver: OpenFOAM ESI v2412 · Mesh: cells: 1218949 · WSS converted from OpenFOAM kinematic units (m^2/s^2) to Pascals by multiplying by rho = 1060.0. Validated against the analytic Poiseuille solution tau = 4*mu*Q/(pi*r^3).

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Zone Analysis and Model Explainability

Case PT-2026-0102



Assessment

Computational fluid dynamics analysis of case PT-2026-0102. Time-averaged wall shear stress at the aneurysm dome measured 0.141 Pa against 2.930 Pa in the parent artery (normalised WSS 0.048), with an oscillatory shear index of 0.000. 92.0% of the sac surface lies below 10% of the parent-artery shear (LSAR). Relative residence time at the dome is 21.35 Pa⁻¹. Sac morphology measured from the reconstructed surface: maximum diameter 11.02 mm, neck 8.22 mm, aspect ratio 1.12, volume 665.6 mm³. Composite Risk Index 59/100 (Moderate).